Canonicalization Before Quadratic Audit: A Certificate-Methods Architecture for an Oriented Level-11 Owner Ledger

Liang Wang[1]
[1]School of Artificial Intelligence and Automation,
Huazhong University of Science and Technology,
Luoyu Road 1037, 430070, Hubei, P.R. China
wangliang.f@gmail.com

2 September 2026

Abstract

This article revises the research architecture for a finite ownership problem associated with a positive time change of the `Gamma_0(11)` geodesic flow. The frozen population contains 138 Hecke-output instances in 55 source-word/prime groups, with oriented primitive `Gamma_0(11)` conjugacy classes as the proposed owners. The earlier design made 9,453 terminal pair dispositions appear to be the foundational certificate. Adversarial review identified that requirement as an unjustified architectural lock. Revision 1 instead makes a deterministic canonicalization map and its biconditional the primary target: two rooted, oriented inputs must receive the same canonical owner bytes exactly when they represent the same oriented primitive owner. The 9,453-row table is retained, but only as a derived adversarial audit of the canonical partition. Aggregate class counts remain weaker post-closure controls. The article also defines the distinct roles of a global owner table `G`, the 138-row incidence relation `I`, and the cell-local no-double-credit quotient `C`. The executed work is limited to closed-corpus evidence synthesis and review-informed method design. No canonicalization theorem, owner partition, all-pairs audit, or `G/I/C` table has been produced. All 22 inherited citations retain `anchor:none`; source identities and metadata close, but claim-to-passage faithfulness remains `INCONCLUSIVE`. The contribution is therefore a reproducible, fail-closed certificate architecture rather than a scientific ownership result. P31 remains A1-only, with no formal Route-A tuple, positive arithmetic A2 credit, or Route-B invocation.

繁體中文摘要

本文針對固定的 $\Gamma_0(11)$ 測地流正時間變換， 提出一套失敗即關閉的 所有權憑證方法架構。 凍結輸入包含 一百三十八個 Hecke 輸出實例、 五十五個來源字與質數群組； 預定所有者是 保持方向的本原 $\Gamma_0(11)$ 共軛類。 核心目標不是先製作 九千四百五十三筆 彼此獨立的成對判定， 而是建立可重播的 決定性規範化映射及其雙條件： 兩個已取本原根 且保留方向的輸入 具有相同規範位元， 當且僅當它們代表 同一所有者。 只有在此定理、 證據格式與獨立驗證器完成後， 全部成對表格 才可作為派生的 對抗式回歸稽核。 本文並嚴格區分 全域所有者表 G、 一百三十八列的關聯表 I 與儲存格內去重商 C。 本階段僅執行 封閉語料的文獻綜合 與審查後方法設計， 未實作規範化器、 共軛判定、 本原根證書、 所有者分割 或任何 $G/I/C$ 結果。 所有引文均無段落定位， 逐主張的來源支持仍屬未定； 本文不宣稱新定理、 普查結果、 新穎性或路線晉級。

Keywords: canonical forms; modular geodesic flow; $\Gamma_0(11)$; oriented conjugacy; primitive owners; certificate methods; adversarial audit

中文關鍵詞：規範形式；模群測地流；保持方向共軛；本原所有者；憑證方法；對抗式稽核

1 Introduction, Research Question, and Contribution

Finite orbit ledgers require an equivalence rule before they can support a dynamical interpretation. A repeated row may represent the same primitive owner, a traversal power, an inverse-oriented owner, or a recurrence of one owner in another correspondence cell. P31 freezes those possibilities rather than allowing an outcome-dependent merge. Its inherited dynamical object is the positive time-changed flow `X_geo/rho_epsilon` on `T^1Y_0(11)`. The real level-11 newform differential, positivity interval, Hecke normalization, reciprocal log-zeta convention, and period coordinate `k=2y+z` remain unchanged.

The finite input is unchanged: 138 instances are distributed over 55 source-word/prime groups. The inherited `2/2/134` split and the three 55-group summaries remain instance- or group-level controls, not owner counts. The proposed owner is an oriented primitive conjugacy class in `Gamma_0(11)` represented by a positive-trace determinant-one lift. The inverse policy has three typed branches. A certified self-reciprocal owner keeps one `owner_bytes` value and records `inverse_relation=self_reciprocal` with its subgroup witness. A certified non-self-reciprocal pair keeps two oriented owner values linked by `inverse_relation=inverse_separated`. If neither proposition is certified, `delta` returns an unresolved inverse disposition. Trace, length, homology, or another filter cannot choose among these branches by itself.

The revised research question is:

Can a deterministic, independently replayable canonicalization contract certify the complete oriented primitive-owner partition of the frozen 138-instance population, and, conditional on that closure, induce the distinct global, incidence, and cell-local estimands `G`, `I`, and `C`?

The primary target is a canonicalization biconditional, not 9,453 independently foundational negative certificates. The all-pairs expansion is derived from the prospective byte partition and can replay sorting, collision, binding, inverse-label, traversal-metadata, and other bookkeeping consequences. It is not an independent semantic truth source. Reflexivity belongs to self fixtures, direction sensitivity to ordered reversal fixtures, and transitivity to explicit triples. Detecting a semantic false merge or split requires a separately bound, target-blind adjudicator; no such route was executed or recorded here.

The closest-work comparison now separates four method families. Finite-index subgroup descriptions and computable Fuchsian domains provide representation and reduction ingredients (P31-S01–P31-S06). Ambient, arithmetic, and hyperbolic conjugacy methods provide decision ingredients that still require an exact `Gamma_0(11)` specialization (P31-S07–P31-S20). Proof-carrying code supplies the general producer-evidence/simple-checker pattern [18], while tamper-evident logging supplies append-only membership and consistency proof patterns [6]. Aggregate class-counting neighbors remain post-closure controls (P31-S21–P31-S22). The project synthesis combines these inherited components into a resolved-domain canonicalizer specification, a total disposition, a derived regression audit, and distinct `G/I/C` projections. It is not a new conjugacy theorem, an implemented solver, a priority claim, or an exhaustive novelty claim.

Three distinctions govern the exposition. First, the dynamical owner is not an input row: ownership is assigned only after subgroup conjugacy, orientation, and maximal-root status are resolved. Second, a proof object is not an estimand: canonical bytes and their witnesses justify a partition, whereas `G`, `I`, and `C` summarize different consequences of that partition. Third, an audit is not its own truth source: the all-pairs expansion can challenge a canonicalizer, but it cannot define correctness when the canonicalizer's biconditional is still open. Keeping these levels separate makes a negative or not-evaluable outcome reportable without silently changing the owner definition.

This architecture also fixes the temporal order of evidence. The owner contract, theorem versions, serialization, and failure states must be registered before the 138 rows are classified. The pair table may be generated only after those inputs have independently replayed certificates. Downstream counts may be read only after the pair audit agrees with the canonical partition or records a typed discrepancy. Nothing in this order presumes that the required canonicalizer exists; the lawful endpoint may remain `NOT_EVALUABLE`.

2 Frozen Literature and Theoretical Boundary

2.1 Exact subgroup representations

The frozen literature describes modular-subgroup structure through several exact mathematical interfaces. Subgroup classification, arithmetic-geometric descriptions, special polygons, and two congruence-recognition routes supply foundational finite-index and finite-data contexts [4, 12–14, 17]. A later computational fundamental-domain framework supplies a potential exact Fuchsian-domain interface [23].

These sources support only a bounded synthesis finding: finite, replayable subgroup data are plausible ingredients. They do not bind the exact P31 representation, prove the required canonicalization biconditional, decide any frozen pair, or define owner bytes. Representation, decision, and certification remain distinct obligations.

For the present article, this is a negative-boundary finding rather than a claim of literature absence. The corpus was frozen by a bounded search and has no passage-level locators. We can say that no retained source was admitted as the complete project solver; we cannot say that no such theorem or method exists anywhere. A later implementation would need a source- finalization step that records the exact representation hypotheses and the passages licensing each subroutine. Until then, even a mathematically promising route is an unbound component rather than an available certificate engine.

2.2 Ambient and arithmetic conjugacy components

The second source group gives candidate components for matrix and arithmetic conjugacy. Ideal-class/matrix correspondences and their later refinements supply ambient integral structure, including a relation between hyperbolic integral matrices and ideal classes [15, 21, 24]. Continued-fraction, arithmetic-group, and modern integral-matrix methods enlarge the possible ambient decision toolkit [1, 7, 10, 11].

No source in this group may be promoted into the required oriented `Gamma_0(11)` solver. A valid specialization would still have to preserve determinant one, positive-trace lifting, the congruence condition, orientation, primitive roots, inverse linkage, termination, and independently replayable evidence. Failure of a bounded ambient search would not constitute a negative subgroup-conjugacy certificate.

Ambient-to-subgroup transfer is a particularly important stop surface. Two matrices may be related under an ambient group while the required conjugator fails the $\Gamma_0(11)$ condition, or an ambient procedure may omit the orientation and root policies needed by the ledger. The future dossier must therefore identify the decision group at every step, prove that normalizations remain within it, and serialize the conjugating or obstructing evidence. Merely reporting that a general algorithm terminated would not show that the project-specific owner predicate was evaluated.

2.3 Canonical reduction, roots, inversion, and replay

Canonicalization requires more than an ambient yes/no decision. Modular-geodesic reduction provides relevant coding context [20]. Word-hyperbolic conjugacy algorithms supply individual and batch decision precedents [3, 8]. Centralizers and reversing symmetries clarify why ordinary symmetry and inversion must be typed separately [2]. Pell equations, unit methods, and computational algebraic number theory provide potential arithmetic subroutines and implementation background [5, 16].

The synthesis does not choose among polygon, arithmetic, or hyperbolic-group implementations. It instead specifies the acceptance condition that any route must meet. A route is admissible only if its hypotheses are proved for the frozen marked object, its output is canonical under the orientation policy, its primitive-root and inverse fields are exact, and an independent verifier can replay success and failure evidence. The literature does not supply the project serialization or prove that this combined contract terminates.

2.4 Aggregate counts as post-closure controls

Class-number structure supplies useful aggregate context [19]. A direct formula for primitive hyperbolic class counts in `Gamma_0(N)` is the closest frozen census precedent, and modular conjugacy has also been related to real-quadratic class numbers [9, 22].

The information content of an aggregate count is lower than that of a labeled partition. Compensating false merges and false splits can preserve a total. Class counts can therefore test a closed owner table for consistency, but they cannot choose canonical representatives, certify an individual owner identity, or establish that every input has been resolved.

Proof-carrying code places a proof next to untrusted producer output so that a simpler consumer can check an explicit safety policy [18]; P31 borrows only that evidence-carrying separation. Tamper-evident history trees support compact membership and append-only consistency checks [6]; P31 borrows only the ledger-verification pattern. Neither source supplies the subgroup-conjugacy theorem, canonical owner bytes, inverse policy, or complete negative certificate needed here. Their role is component positioning, not evidence of priority or of a completed P31 method.

3 Executed Methodology

3.1 Closed-corpus literature synthesis

The executed research method remains literature synthesis over frozen artifacts. The upstream corpus captured 44 manifestations, removed nine duplicates, screened 35 unique records, excluded 13, and retained 22 sources. A dated replay on 3 September 2026 submitted all 20 exact frozen query strings to Crossref `query.bibliographic`, bounded to one top-ranked metadata record per query. The row-level supplement records each query, timestamp, interface, HTTP status, candidate identity, inventory match, decision, and reason. It is a new replay, not a reconstruction of the unavailable original-session exclusion rows. A separate bounded closest-work search verified exactly two method records and added them only to the notes-side versioned bibliography.

The method-component matrix has 24 rows: P31-S01–P31-S22 retain `anchor:none` and `INCONCLUSIVE` theorem-passage transfer, while P31-S23 and P31-S24 have finalized publisher-level method locators for the narrow proof-carrying and tamper-evident patterns cited here. Every row states the component or claim role, passage status, applicable hypothesis or scope, and a prohibited transfer. Metadata closure does not promote the 22 unresolved rows, and the two new method rows do not establish any P31 owner theorem, implementation, or semantic adjudicator. No direct quotation is used.

The synthesis uses source-effect discipline rather than vote counting. A source may support subgroup representation, ambient reduction, proof-carrying checking, or append-only ledger verification while being explicitly excluded from proving the composite P31 owner contract. Venue recognition and algorithmic proximity do not erase hypothesis, object, or output-type boundaries. The method-passage matrix records those permitted components and forbidden transfers row by row; missing theorem-to-certificate links remain visible obligations.

Citation closure is checked against the notes-side versioned bibliography: all 24 cited source identifiers resolve to one entry, including source-verified P31-S23 and P31-S24. The canonical bibliography is unchanged. This structural check does not validate a theorem passage. P31-S01–P31-S22 retain `anchor:none` and `INCONCLUSIVE` claim-to-passage status; only the narrow method descriptions for P31-S23/P31-S24 use their finalized publisher-level locators. The wording remains limited to inherited components, prospective use, and explicit exclusions.

3.2 Review-adjudicated revision procedure

The executed scholarly method was a closed-corpus synthesis: records were captured, deduplicated and screened under the frozen inclusion frame; retained sources were coded by admitted contribution, excluded stronger use, and applicability warning; and the resulting limitations were propagated into the design claims. Four procedurally separated, single-model-family roles then reviewed the resulting report. Their `MAJOR_REVISION` outcome and the author's adjudication are development provenance, not evidence for a conjugacy theorem, owner decision, or scientific result. Revision 1 changes only the prospective contract and its stated evidence boundary.

No new retrieval occurred during Revision 1. No source field, reference, passage locator, direct quotation, experiment, code result, proof result, or canonical manuscript byte was introduced. The Phase-6 ClaimIntent manifest was frozen before this prose and supplies the complete set of eight substantive article claims. Every finding below is either a closed-corpus evidence-synthesis statement, a project definition, or a prospective method obligation.

4 Review-Adjudicated Certificate Architecture

4.1 Primary target: a canonicalization biconditional

Let `X` denote the frozen set of 138 instances after exact input validation. The future method should define a partial operation `root(x)` that either returns an oriented primitive `Gamma_0(11)` representative together with a traversal exponent and proof payload, or returns a typed not-evaluable state. It should then define a deterministic byte map

```
OwnerDisposition = Resolved(owner_bytes, witnesses)
                 | Unresolved(code, evidence)
delta: X -> OwnerDisposition
X_res = {x in X: delta(x) is Resolved}
kappa: X_res -> OwnerBytes
Closed = (X_res = X)
```

Thus `delta` is total even when it returns a typed unresolved disposition; `kappa` is defined only on the resolved domain. A total owner map on all of `X` exists only after the separately checked condition `Closed` holds.

on every successfully resolved input. The primary theorem target is the biconditional

```
kappa(x)=kappa(y)
if and only if
root(x) and root(y) represent the same oriented primitive Gamma_0(11) owner.
```

The inverse rule is branch-complete but remains prospective. If an exact `Gamma_0(11)` witness certifies that an oriented primitive owner is conjugate to its inverse, one canonical value is retained and the certificate records `inverse_relation=self_reciprocal`, the conjugator, presentation version, and replay trace. If a lawful obstruction certifies nonconjugacy, two oriented values are retained and linked as `inverse_separated`. If neither witness exists, `delta` returns `UNRESOLVED_INVERSE_RELATION`; it must not choose two owners merely from a convention. This typed branch does not assert an exclusion lemma or report that any frozen instance is self-reciprocal.

On the resolved domain, the certificate invariant is

$$\forall x, y \in X_{\text{res}}, \qquad \kappa(x) = \kappa(y) \iff x \text{ and } y \text{ represent the same oriented primitive owner.}$$

Soundness, completeness, and determinism are therefore claims about `kappa:X_res->OwnerBytes`. They become a total-owner theorem for the frozen population only after the independent closure condition `X_res=X`. Before closure, byte equality can induce a partition of the resolved subset but cannot certify a complete partition of `X`.

The branches have distinct failure modes. Soundness fails if distinct resolved oriented owners share bytes; completeness fails if one resolved owner receives different bytes; and determinism fails if a resolved input can receive different bytes under permitted executions. `delta` itself remains total because it may emit a typed `Unresolved` value. Global owner-map closure fails whenever `X_res` is a proper subset of `X`; in that state `kappa` is not total on `X`, the all-population biconditional is not available, and complete `G/I/C` materialization is prohibited.

The primitive-root layer must precede canonical owner serialization. If an input is a traversal power, serializing the unreduced input could mint a false owner. Conversely, root extraction must not erase the traversal exponent needed by the incidence relation. The proposed certificate therefore carries both the primitive owner bytes and the exact repetition field. Hecke degree remains an input coordinate and is never substituted for that repetition field.

4.2 Prospective certificate and verifier contract

Each future per-instance certificate binds the immutable input identifier and hash; subgroup presentation and theorem version; exact membership evidence; normalized matrix or word; maximal primitive root and traversal exponent; orientation convention; canonical owner bytes; and the proof or reduction trace for a read-only verifier. It also carries `inverse_relation` in {`self_reciprocal`, `inverse_separated`, `unresolved`}, with the matching subgroup conjugator, obstruction, or unresolved evidence. A failure record identifies the exact failed precondition or theorem obligation; a timeout never becomes nonconjugacy.

The verifier acceptance predicates are also prospective. It should recheck input binding, determinant and subgroup membership, root powering, primitiveness under the chosen theorem, orientation preservation, deterministic serialization, and any inverse relation. It should reject unknown fields, stale theorem versions, noncanonical encodings, missing proof data, or an unresolved subroutine. Independent replay means an implementation separate from the producer can evaluate the frozen mathematical predicates; it does not mean that two AI reviews have statistically independent errors.

This is a non-executable design specification, not an implemented contract. No schema bytes, theorem registry, frozen fixture corpus, producer, independent verifier, or build manifest exists in the recorded project state. Accordingly, reproducibility in this article applies only to the literature and review artifacts listed in the materials statement; it does not apply to an owner result, a scientific replay, or an executable certificate pipeline. Creating those assets requires a separately authorized implementation stage.

The producer and verifier should communicate through a versioned, closed schema. Required fields should have fixed encodings, and unknown fields should cause rejection rather than permissive parsing. Each proof payload should state which theorem and representation version it uses, so that a change in a normal-form convention cannot silently reuse old bytes. A manifest should bind the accepted input hashes, fixture hashes, producer build, verifier build, and output hashes. These requirements describe a prospective reproducibility contract; no such manifest or build exists in the present article.

Positive and negative evidence also require asymmetric handling. A positive conjugacy certificate can carry an explicit conjugator whose membership and action are replayed. A negative disposition needs a theorem-backed exhaustive argument appropriate to the frozen group and representation; failure to find a conjugator is insufficient. Root certificates face the same distinction: displaying a power relation is positive evidence, while claiming no proper root requires an exhaustive licensed method. The verifier must preserve these differences instead of reducing every result to one Boolean field.

Target-blind fixtures should exercise the contract before population execution. At minimum they should include identical inputs, distinct representatives of one owner, inverse-related owners, proper powers, coarse-invariant collisions, malformed membership data, and deliberately unresolved theorem bindings. Expected outputs must be registered before the producer sees them. A fixture suite can reveal implementation defects, but passing it cannot replace the global biconditional or the complete 138-instance replay.

4.3 The 9,453-row table as a derived adversarial audit

For `|X|=138`, the 9,453 unordered distinct-pair rows are generated from the same prospective canonical-byte partition. They can replay byte equality, sorting, binding, inverse labels, traversal metadata, and bookkeeping consequences. They cannot test reflexivity, ordered reversal, or three-input closure, and they cannot detect semantic false merges, false splits, or nontransitivity without expected dispositions from a separately bound, target-blind adjudicator. No such independent semantic route exists in the recorded project, so the optional direct-solver field remains `ABSENT` and the all-pairs surface is not a truth source.

Prospective coverage is separated by fixture type. Self fixtures `F_self={(x,x)}` test reflexive handling; ordered reversal pairs test direction-sensitive serialization; triples `(x,y,z)` test closure and transitivity consequences; and separately sourced disagreement rows would compare canonical bytes with a target-blind semantic adjudicator. The 9,453 unordered distinct pairs cover none of the first three fixture shapes by construction and have no independent expected semantic labels. They are limited to byte and bookkeeping checks. No fixture set or direct route was executed.

The table is not described as a uniquely necessary proof architecture. If a future scientific requirement needs a separately replayable negative obstruction for particular cross-class pairs, that need must be justified and bound. Until then, the full table is a regression expansion of the canonical certificate rather than 9,453 independent foundations.

The audit should retain discrepancies rather than overwrite them. If a direct pair solver and canonical-byte equality disagree, the row should record both results, their proof payloads, and a fail-closed batch state. Neither route should be selected after inspecting which one preserves a preferred aggregate count. This makes the table an adversarial consumer of the canonical partition rather than a mechanism for repairing the partition post hoc.

Pair rows can also test equivalence-relation consequences globally. A same-owner relation derived from canonical bytes is automatically reflexive, symmetric, and transitive at the byte level, but producer or serialization defects may still appear as inconsistent input bindings, inverse labels, or traversal metadata. The 9,453-row expansion exposes all unordered cross-input comparisons and therefore remains the most demanding finite regression surface even though it is not the primary proof object.

4.4 Distinct `G`, `I`, and `C` estimands

The downstream relations are conditional on the total disposition `delta`. Complete `G/I/C` materialization requires `X_res=X`. Until then, unresolved rows enter only `I_diag`, contribute to no estimand, and prevent `G`, `I`, or `C` from being published as complete. Once closure holds, `I` contains exactly 138 provenance-preserving input rows; `G` is its distinct owner projection; and `C` is its distinct cell-owner projection. The consolidated schema below is the checkable contract; the surrounding prose supplies interpretation only.

```
G = {kappa(x): x in X},
```

with exactly one row per distinct oriented owner byte string. Define the incidence relation `I` with one row for every input instance, carrying its input fields, cell coordinates, owner bytes, traversal exponent, inverse link, and all frozen Hecke coordinates. Thus `|I|=138` even when `|G|<138`.

Let a cell key be the frozen tuple `(source_word, prime, hecke_degree)`. Define `C` as the set of distinct `(cell_key, owner_bytes)` pairs induced by `I`. Raw multiplicity remains visible in `I`, while `C` awards one unit to an owner within a cell. The same owner may appear once in each of several cells without being duplicated in `G`.

Conditional well-definedness is straightforward but important. If `kappa` is total and satisfies the biconditional, changing an input representative within the same oriented owner cannot change its `G` row or its cell-level owner key. Deduplication inside a cell is therefore invariant under representative choice. Conversely, two different canonical owner bytes cannot be merged in `C` without violating the definition. This set-theoretic argument does not prove that `kappa` exists or that the frozen inputs are resolved; it states what follows if the primary certificate closes.

The three estimands answer noninterchangeable questions. The cardinality of `G` concerns distinct global oriented owners represented in the population. The rows of `I` retain provenance and allow one owner to be traced back to every input occurrence. The rows of `C` enforce no-double-credit within a frozen correspondence cell while allowing the same owner to occur in several cells. Publishing only `G` or only `C` would destroy occurrence-level information needed to reconstruct `I`; conversely, a complete `I` can induce `G` and `C` by the stated projections, but separate materializations make the transformations independently auditable. For that reason their schemas, validation rules, and summary statistics should be declared independently.

Relation	Key	Closed cardinality	Materialization gate	Provenance and allowed direction
`G`	`owner_bytes`	distinct resolved owners	`X_res=X`	derived from complete `I`; cannot reconstruct input occurrences
`I`	`input_id`, foreign key to `G`	138	`X_res=X`	retains input, cell, traversal, Hecke, hash, and disposition provenance; may project to `G` and `C`
`C`	`(cell_key,owner_bytes)`	distinct complete-`I` cell-owner pairs	`X_res=X`	derived from complete `I`; cannot reconstruct input occurrences
`I_diag`	`input_id`	all unresolved diagnostics	any unresolved row	diagnostic only; no projection into complete estimands

Therefore the only allowed construction direction is complete `I -> G,C`; neither `G` nor `C`, separately or jointly, may be used to reconstruct `I`. The three published estimands retain separate schemas and validation rules.

Materialization must remain conditional on zero unresolved owner rows. If even one input cannot be rooted or canonicalized, `I` may retain a diagnostic record but `G` and `C` cannot be reported as complete estimands. A partial table may be useful for method development only if it is labeled incomplete and does not award owner counts. This stop rule prevents missing inputs from being silently treated as new, different, or irrelevant owners.

5 Evidence-Synthesis Findings

Eight precommitted positions control this manuscript. First, the frozen corpus supplies complementary subgroup, ambient-conjugacy, arithmetic, reduction, and hyperbolic-group components, but no executed complete oriented `Gamma_0(11)` owner solver. Second, the primary certificate target is the deterministic canonicalization biconditional. Third, the 9,453-row table is a derived adversarial audit, and aggregate counts remain post-closure controls. Fourth, `G`, `I`, and `C` are distinct estimands populated only after owner closure.

Fifth, a complete prospective certificate preserves subgroup membership, orientation, primitive-root status, inverse linkage, traversal exponent, deterministic serialization, and replayable failure evidence while keeping Hecke degree separate. Sixth, the article's methodological contribution is the separation of canonical proof object, audit expansion, and estimands. Seventh, metadata and source-identity closure do not clear theorem passages, retraction status, source conflicts, or the theorem-to-certificate bridge. Eighth, the manuscript remains A1-only and changes no scientific result or Route state.

These are evidence-synthesis and design findings. They do not imply feasibility, novelty, theorem correctness, or implementation readiness. The central scientific state remains `NOT_EXECUTED`.

6 Reproducibility and Prospective Interface

Reader-accessible Stage-4-prime materials are exactly the entries listed in `notes/stage4_prime_reader_artifact_manifest_round2.json` (SHA-256 `0b2d2692f93eccd104183aa45381ac461f310c3fdc912906b225ed26cc16be00`). Each row names its schema or format version, byte length, full SHA-256 digest, and repository-relative access state. The branch-relative locator is `https://github.com/maris205/hilbert-polya-structure/tree/main/flow_systems/papers/31-level11-conjugacy-owner-ledger`; it is not a content-addressed or persistent archive. Recovery claims are made only for listed entries. Generative prose is not promised byte-reproducible, and no owner ledger or scientific result package is listed.

A later scientific package should freeze four layers before reading owner outcomes. Layer one binds the exact `Gamma_0(11)` representation and proves every input conversion. Layer two binds the canonicalization and root theorem, including both directions of the biconditional. Layer three freezes producer and verifier schemas, adversarial fixtures, and typed failure states. Layer four defines the derived all-pairs audit and the conditional `G/I/C` materialization. Hashes, theorem versions, serialization versions, and fixture manifests should be immutable across execution.

The smallest valid next test is not the census. It is a theorem-to-certificate dossier evaluated against deliberately small, target-blind fixtures. Scientific population execution requires separate authorization after the primary biconditional, root policy, inverse policy, and independent verifier have closed. A missing theorem or unresolved fixture should terminate as `NOT_EVALUABLE_CONJUGACY_INCOMPLETE`, not as a negative owner result.

7 Discussion and Implications

The revised architecture improves the match between the mathematical object and its evidence. A partition is naturally represented by a total equivalence invariant. Expanding that invariant to every pair can be an excellent audit, but expansion size should not be confused with proof content. This distinction reduces a false quadratic necessity while preserving the most demanding adversarial check available for the fixed population.

The change does not make P31 easy. The canonicalization theorem must simultaneously handle exact subgroup membership, oriented conjugacy, primitive roots, inverse linkage, and deterministic bytes. A compact certificate that lacks either direction of the biconditional would be weaker than the all-pairs design it replaces. The revision therefore narrows the foundational object without lowering the exactness standard.

The `G/I/C` separation also clarifies what a future result would mean. `G` answers how many global oriented primitive owners occur. `I` records how the 138 frozen instances map to those owners. `C` answers how owner identity contributes within each frozen correspondence cell. None of the inherited instance summaries can substitute for these estimands, and no cell-local recurrence can multiply the number of global owners.

The architecture is intentionally compatible with more than one future mathematical implementation. A polygonal reducer, an arithmetic route, or a word-hyperbolic route could be considered, provided its exact hypotheses and evidence are bound to the same owner semantics. This interoperability is conditional: different producers cannot be compared until they preserve orientation, root status, inverse linkage, and serialization under one independently checked contract. The article therefore standardizes the meaning of a prospective result without claiming that any implementation route is presently adequate.

Synthetic interoperability trace (not executed). A hypothetical polygon reducer and a hypothetical word-hyperbolic reducer may use route-private proof tags, but each proposed envelope must expose the same input digest, membership disposition, primitive root, traversal exponent, orientation bit, inverse link, owner-schema version, and payload digest. A read-only verifier would reject a missing common field and would return `UNRESOLVED_SCHEMA_MISMATCH` when the versions are incompatible. This illustration contains no real input, owner byte string, solver output, or scientific comparison.

For Route A, this remains primitive-owner infrastructure at A1. No primitive Euler product, dynamical determinant, zero comparison, analytic continuation, or arithmetic specificity test is produced. Editorial improvement cannot promote a Route coordinate. Route B remains closed because there is no Route-A readiness, Hilbert space, operator, domain, self-adjointness result, trace formula, or divisor identity.

8 Acknowledged Limitations

The dated supplement exposes one screening decision for each of the 20 frozen query strings, but it is a new bounded Crossref replay and not a reconstruction of the unavailable original-session excluded rows. The 24-row method-passage matrix preserves `INCONCLUSIVE` theorem transfer for P31-S01–P31-S22 and finalizes only the narrow publisher-level method locators for P31-S23 and P31-S24. Those two records position proof-carrying and tamper-evident components; they do not provide a `Gamma_0(11)` owner theorem, solver, inverse witness, or semantic adjudicator. Retraction and source-conflict screening were not expanded by this revision.

The corpus has no general current retraction or source-conflict clearance. Those checks remain unrun and their status remains unknown. The P31-S16 page correction is preserved, but one corrected field does not constitute a clean integrity screen. The corpus is historically weighted and no new contribution or novelty comparison was authorized.

The method architecture is uninstantiated. There is no bound canonicalization theorem, no producer or independent verifier, no completed fixture suite, no owner decision, and no `G/I/C` output. The conditional definitions show how outputs would relate after closure; they do not close the primary mathematical obligation.

Finally, AI-assisted review and drafting used one Codex model family with procedural role separation. This supplies multiple documented perspectives, not statistically independent validation. Liang Wang's gate confirmations establish author decisions and workflow authority only; they are not evidence of personal full-text checking.

9 Future Work

Future work should proceed in this order:

1. perform a separately authorized source-finalization pass that freezes exact theorem passages, hypotheses, correction status, and any required source-conflict or retraction checks;
2. bind one exact subgroup representation and prove the conversion from every frozen input;
3. state and prove the canonicalization biconditional, maximal-root contract, orientation convention, and inverse-link rule;
4. freeze certificate schemas, producer/verifier roles, typed failures, and target-blind adversarial fixtures;
5. execute the 138 per-instance certifications only after the interface recheck passes;
6. derive the 9,453-row audit and compare it with optional direct pair checks without treating aggregate counts as pair evidence; and
7. materialize and independently validate `G`, `I`, and `C` only after zero unresolved owner inputs.

Each step requires a new gate when it extends beyond literature-only manuscript composition. Nothing in this article authorizes retrieval or scientific execution.

10 Conclusion

P31 now has a more defensible certificate architecture. The foundational target is a total deterministic canonical owner map with a proved biconditional for oriented primitive `Gamma_0(11)` ownership. The 9,453-row table remains a full-population adversarial audit derived from that certificate, not a presumed uniquely necessary collection of bespoke proofs. Aggregate class counts remain weaker controls, and the global owner table `G`, incidence relation `I`, and cell-local quotient `C` remain distinct conditional outputs.

This is concrete design-level progress, but it is not a scientific result. No canonical form, owner partition, pair audit, theorem, computation, or estimand table has been executed. The source corpus remains passage-unresolved. A bounded Stage 2.5 textual-originality screen found no exact match within its declared samples and corpora, but scientific contribution novelty remains unassessed; the formal Route-A tuple remains `UNASSIGNED`, positive arithmetic A2 remains absent, and Route B remains closed.

Declarations

Author contribution and accountability. Liang Wang specified the mathematical object and restrictions, supplied the conceptual direction, approved the recorded stage gates, adjudicated the certificate-first design, and is responsible for the manuscript's scholarly content. AI assistance is disclosed below and is not credited with authorship. Stage-gate approval does not attest that the author personally performed full-text or exact source-passage verification.

Funding. No funding was received for this work.

Competing interests. The author declares no competing interests.

Ethics statement. Human-subjects and animal-research review is not applicable. The work concerns theoretical mathematics, published-source records, and project-owned workflow artifacts; it involves no participants, identifiable personal data, animals, recruitment, or intervention.

Data and materials availability. The bounded reader-accessible files are enumerated in `notes/stage4_prime_reader_artifact_manifest_round2.json` (SHA-256 `0b2d2692f93eccd1 04183aa45381ac461f310c3fdc912906b225ed26cc16be00`), with a schema or format, byte length, full digest, and repository-relative access state for every entry. The notes-side versioned bibliography includes source-verified P31-S23 and P31-S24; the canonical bibliography remains frozen. The branch-relative repository locator is not a persistent archival identifier, and materials outside the manifest are not described as reader-recoverable. No owner ledger, scientific dataset, solver output, pair-decision table, or `G/I/C` result was generated.

AI Disclosure and Verification Limitation

OpenAI Codex, using the GPT-5 model family, assisted during the session dated 2026-09-02 UTC; the exact backend snapshot/build was not exposed. AI-assisted work in the recorded pipeline included literature-search support, source-identity and metadata checking, evidence-matrix construction, evidence synthesis, report drafting, four role-based Phase-5 reviews, review synthesis, ClaimIntent-constrained Revision-1 drafting, citation-ID/reference closure checks, and revision-log accounting. No AI system executed a P31 solver, proof, pair decision, owner census, experiment, or canonical-results refresh.

Liang Wang is the responsible human author. He approved the project restrictions, stage gates, and the Phase-6 author-adjudicated design choice. Those approvals must not be interpreted as a statement that he personally read every source in full or verified any claim at the exact source-passage level. The recorded verification was bounded mainly to source identity, metadata, abstracts, authoritative landing pages, and project-local claim-fitness records. All 22 citations lack passage locators, so claim-to-passage faithfulness remains `INCONCLUSIVE`; the article does not claim theorem-level source verification, novelty clearance, or a clean retraction/conflict screen.

References

[1] Harry Appelgate and Hironori Onishi. Continued fractions and the conjugacy problem in $SL_2(\mathbb{Z})$. *Communications in Algebra*, 9(11):1121–1130, 1981. doi: 10.1080/00927878108822637.

[2] Michael Baake and John A. G. Roberts. Symmetries and reversing symmetries of toral automorphisms. *Nonlinearity*, 14(4):R1–R24, 2001. doi: 10.1088/0951-7715/14/4/201.

[3] David J. Buckley and Derek F. Holt. The conjugacy problem in hyperbolic groups for finite lists of group elements. *International Journal of Algebra and Computation*, 23(5):1127–1150, 2013. doi: 10.1142/S0218196713500203.

[4] Shih-Ping Chan, Mong-Lung Lang, Chong-Hai Lim, and Ser-Peow Tan. Special polygons for subgroups of the modular group and applications. *International Journal of Mathematics*, 4(1):11–34, 1993. doi: 10.1142/S0129167X93000030.

[5] Henri Cohen. *A Course in Computational Algebraic Number Theory*, volume 138 of *Graduate Texts in Mathematics*. Springer, 1993. doi: 10.1007/978-3-662-02945-9.

[6] Scott A. Crosby and Dan S. Wallach. Efficient data structures for tamper-evident logging. In *18th USENIX Security Symposium (USENIX Security 09)*, Montreal, Quebec, 2009. USENIX Association. URL `https://www.usenix.org/conference/usenixsecurity09/technical-sessions/presentation/efficient-data-structures-tamper-evident`.

[7] Bettina Eick, Tommy Hofmann, and E. A. O'Brien. The conjugacy problem in $GL(n, \mathbb{Z})$. *Journal of the London Mathematical Society*, 100(3):731–756, 2019. doi: 10.1112/jlms.12246.

[8] David B. A. Epstein and Derek F. Holt. The linearity of the conjugacy problem in word-hyperbolic groups. *International Journal of Algebra and Computation*, 16(2):287–305, 2006. doi: 10.1142/S0218196706002986.

[9] V. V. Golovchanskii and M. N. Smotrov. An explicit formula for the number of classes of primitive hyperbolic elements in the group $\Gamma_0(N)$. *Sbornik: Mathematics*, 199(7):1009–1031, 2008. doi: 10.1070/SM2008v199n07ABEH003951. URL `https://www.mathnet.ru/eng/sm3853`.

[10] Fritz Grunewald and Daniel Segal. Some general algorithms. I: Arithmetic groups. *Annals of Mathematics*, 112(3):531–583, 1980. doi: 10.2307/1971091.

[11] Fritz J. Grunewald. Solution of the conjugacy problem in certain arithmetic groups. In *Word Problems II*, volume 95 of *Studies in Logic and the Foundations of Mathematics*, pages 101–139. North-Holland, 1980. doi: 10.1016/S0049-237X(08)71335-1.

[12] Tim Hsu. Identifying congruence subgroups of the modular group. *Proceedings of the American Mathematical Society*, 124(5):1351–1359, 1996. doi: 10.1090/S0002-9939-96-03496-X.

[13] Ravi S. Kulkarni. An arithmetic-geometric method in the study of the subgroups of the modular group. *American Journal of Mathematics*, 113(6):1053–1133, 1991. doi: 10.2307/2374900.

[14] Mong-Lung Lang, Chong-Hai Lim, and Ser-Peow Tan. An algorithm for determining if a subgroup of the modular group is congruence. *Journal of the London Mathematical Society*, 51(3):491–502, 1995. doi: 10.1112/jlms/51.3.491.

[15] C. G. Latimer and C. C. MacDuffee. A correspondence between classes of ideals and classes of matrices. *Annals of Mathematics*, 34(2):313–316, 1933. doi: 10.2307/1968204.

[16] Jr. Lenstra, H. W. Solving the Pell equation. *Notices of the American Mathematical Society*, 49(2): 182–192, 2002. URL `https://www.ams.org/notices/200202/fea-lenstra.pdf`.

[17] M. H. Millington. Subgroups of the classical modular group. *Journal of the London Mathematical Society*, s2-1(1):351–357, 1969. doi: 10.1112/jlms/s2-1.1.351.

[18] George C. Necula. Proof-carrying code. In *Proceedings of the 24th ACM SIGPLAN-SIGACT Symposium on Principles of Programming Languages*, pages 106–119. ACM Press, 1997. doi: 10.1145/263699.263712. URL `https://doi.org/10.1145/263699.263712`.

[19] Peter Sarnak. Class numbers of indefinite binary quadratic forms. *Journal of Number Theory*, 15(2): 229–247, 1982. doi: 10.1016/0022-314X(82)90028-2.

[20] Caroline Series. The modular surface and continued fractions. *Journal of the London Mathematical Society*, s2-31(1):69–80, 1985. doi: 10.1112/jlms/s2-31.1.69.

[21] Olga Taussky. On a theorem of Latimer and MacDuffee. *Canadian Journal of Mathematics*, 1(3): 300–302, 1949. doi: 10.4153/CJM-1949-026-1.

[22] Charles Traina. The conjugacy problem of the modular group and the class number of real quadratic number fields. *Journal of Number Theory*, 21(2):176–184, 1985. doi: 10.1016/0022-314X(85)90049-6.

[23] John Voight. Computing fundamental domains for Fuchsian groups. *Journal de Théorie des Nombres de Bordeaux*, 21(2):467–489, 2009. doi: 10.5802/jtnb.683.

[24] D. I. Wallace. Conjugacy classes of hyperbolic matrices in $SL(n, \mathbb{Z})$ and ideal classes in an order. *Transactions of the American Mathematical Society*, 283(1):177–184, 1984. doi: 10.1090/S0002-9947-1984-0735415-0.